Topic: 8086 CPU Architecture

Q. Explain the internal architecture of the 8086 microprocessor. Describe the functions of the Bus Interface Unit (BIU) and the Execution Unit (EU).

> 📌 **Definition to Remember**
> The **Intel 8086** is a 16-bit microprocessor with a 20-bit address bus (1 MB memory). Its architecture is divided into two asynchronously operating units: the **Bus Interface Unit (BIU)** (handles memory/bus fetches) and the **Execution Unit (EU)** (decodes and executes instructions), enabling early instruction pipelining.

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1. Bus Interface Unit (BIU)

The BIU handles all data and address transfers on the system buses.
* **Functions:** Calculates physical addresses, fetches instructions from memory, reads/writes data.
* **Components:**
* **Segment Registers:** `CS`, `DS`, `SS`, `ES` (for memory segmentation).
* **Instruction Pointer (IP):** Holds the offset of the next instruction.
* **Address Generation Circuit:** Calculates `Physical Address = (Segment * 10H) + Offset`.
* **Instruction Queue:** A 6-byte FIFO queue. Prefetches up to 6 bytes of instructions to keep the EU busy (Pipelining).

2. Execution Unit (EU)

The EU is responsible for decoding and executing the fetched instructions.
* **Functions:** Pulls instructions from the BIU's queue, decodes them, performs operations, and requests BIU to store results.
* **Components:**
* **ALU:** 16-bit Arithmetic Logic Unit.
* **General Purpose Registers:** `AX`, `BX`, `CX`, `DX`.
* **Pointer/Index Registers:** `SP`, `BP`, `SI`, `DI`.
* **Flag Register:** 16-bit register showing ALU status.
* **Control Circuitry:** Decodes and generates control signals.

3. Parallel Operation

Because the BIU and EU operate independently, the BIU continuously fills the 6-byte queue from memory while the EU executes the previously fetched instructions. This prevents the EU from sitting idle waiting for memory fetches, drastically reducing execution time.

> ⭐ **Must-Write Points (for 10 marks)**
> 1. 8086 architecture is divided into the **BIU (Bus Interface Unit)** and **EU (Execution Unit)**.
> 2. **BIU** handles memory/bus operations and calculates the 20-bit physical address.
> 3. BIU contains Segment Registers (CS, DS, SS, ES) and a **6-byte Instruction Queue** for pipelining.

4. **EU** decodes and executes instructions from the queue.
5. EU contains the ALU, General Purpose Registers (AX, BX, CX, DX), Pointers, and Flags.
6. The independent operation allows the BIU to prefetch instructions while the EU executes them.
7. This parallel operation minimizes CPU idle time.

Quick Recall

`8086 Arch → BIU + EU → BIU (Bus transfers, Address Calc, 6-byte Queue, Seg Registers) → EU (Execute, ALU, Gen Registers, Flags) → Parallel operation (Pipelining)`